Mitigration :

Migration means **moving data, applications, or systems** from **one place to another** safely and correctly.
Example:

- From **PostgreSQL** → **Snowflake**
- From **On-premise** → **Cloud**
- From **Old system** → **New system**

Migration ensures:

- No data loss
- Minimal downtime
- Correct structure & performance

There are two main topics in Migration :

1. Tools 2. Guides

1. Tools : They are **software or utilities** used to perform the migration work.

Purpose of Tools :

- Extract data from source
- Transform data if needed
- Load data into target system
- Validate migrated data

Common Migration Tool Activities:

- Data copy
- Schema conversion
- Data validation
- Error handling

Examples:

- Snowflake Migration Tools
- ETL tools (Informatica, Talend)
- Database utilities
- Scripts (SQL, Python.

In short : Tools How the migration is done.

In this, they are two main tools:

- Snowconvert Ai
- Snowpark Migration Accelerator

What is SnowConvert AI?

- **SnowConvert AI** is a tool that **automatically converts code from other databases into Snowflake code**.
- It translates old database language into Snowflake language

Why SnowConvert AI is Used

- To **move from legacy databases to Snowflake**
- To **save manual coding effort**
- To **reduce mistakes**
- To **speed up migration**

What It Converts : It converts code from:

- Oracle
- Teradata
- SQL Server
- Netezza into **Snowflake SQL**

What is a Snowpark Migration Accelerator?

- Snowpark Migration Accelerator helps move application logic (code) into Snowflake using Snowpark.
- It helps run your application logic inside Snowflake

Why Snowpark Migration Accelerator is Used

- To move business logic into Snowflake
- To reduce data movement
- To improve performance
- To modernize applications

What It Migrates

- Python
- Java
- Scala
- Spark-based logic into Snowflake Snowpark

Under the snowpark migration these are :

User Guide

What is a User Guide?

The User Guide tells you how to use the tool step by step.

- Installation
- Setup
- Commands
- How to run migration

Use Cases

What are Use Cases?

Use cases explain real-life situations where this tool is useful.

- Migrating Spark jobs to Snowflake
- Moving Python logic into Snowflake
- Reducing external servers

Issue Analysis

What is Issue Analysis?

- Find problems in your existing code
- Understand what cannot be migrated easily
- Fix errors before migration

Workspace Estimator

What is Workspace Estimator ? It helps estimate:

- How much effort
- How much work
- How complex the migration will be

Interactive Assessment Application

- Scans your code
- Gives a migration readiness report

- Shows how much can be automated

Support

What is Support?

Support provides:

- Help documents
- Troubleshooting
- Contact options

1. General Troubleshooting

This is **basic problem-fixing help**.
If something is not working, this section helps you:

- Find what went wrong
- Try simple fixes
- Solve common issues step by step

Example : If the app is slow, restart it" or "Check your internet connection.

2. Frequently Asked Questions (FAQs)

This section answers **common questions** that many users ask again and again.
It helps you get quick answers without contacting support.

Example:

- How do I reset my password?
- Why can't I log in?

2. Guides : Guides are step-by-step instructions or documentation that explain how to use the tools and follow best practices.

Purpose of Guides:

- Explain migration steps
- Define best practices
- Avoid common mistakes
- Ensure correct setup

What Guides Usually Contain:

- Prerequisites
- Step-by-step process
- Architecture diagrams
- Validation & testing steps
- Troubleshooting tips

Examples:

- Snowflake Migration Guide
- Best practices document
- User manuals
- Setup instructions

In short : Guides = How to use the tools correctly.

```
Migration
|
├── Tools
|     └── Software used to move data
|
└── Guides
      └── Instructions on how to migrate
```

1. Teradata

A guide that explains **how to work with Teradata** (a data warehouse).
It helps users understand:

- How to connect
- How to run queries
- How to move data from Teradata to another system

Example : How to extract data from Teradata.

2. Databricks

This guide shows **how to use Databricks**, which is used for **big data and analytics**.
It helps users:

- Run data processing jobs
- Use notebooks
- Integrate Databricks with other tools

Example : How to load and analyze data in Databricks.

3. SQL Server

This guide helps users who work with **Microsoft SQL Server**.
It explains:

- How to connect to SQL Server
- How to write SQL queries
- How to migrate data safely

Example : How to move data from SQL Server.

4. Amazon Redshift

A guide for **Amazon's cloud data warehouse**.
It helps users:

- Load data into Redshift
- Query data
- Optimize performance

Example : How to migrate data from Amazon Redshift.

5. Oracle

This guide explains how to work with **Oracle databases**.
It helps users:

- Extract data
- Convert Oracle data
- Move Oracle data to modern platforms

Example : Oracle to cloud migration steps.

6. Azure Synapse

A guide for Microsoft Azure's analytics service.
It explains:

- How to analyze large data
- How to connect Synapse to other tools
- How to migrate data

Example : Using Azure Synapse for reporting.